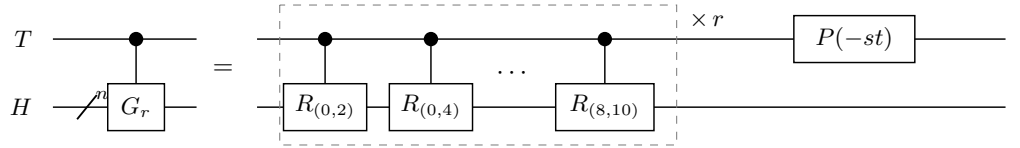
(a) Controlled evolution on the hit register, c-G_r, for $A_1 = sI - A_{\text{hit}}$



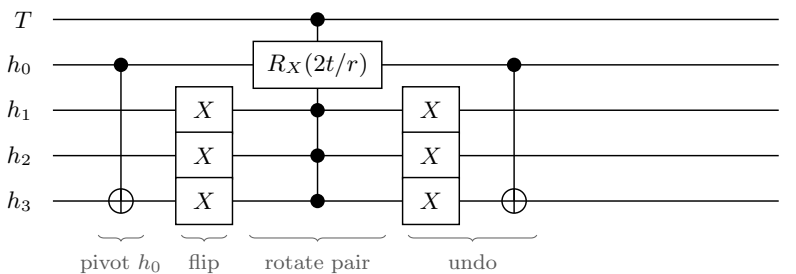
$s = 4$; a proposal (7.10), not in the package

$R_{(i,j)} = e^{-i(t/r)X_{ij}}$, one per edge of A_{hit} (worked event: 20 edges, two K_5)

$G_r = e^{-ist}\Big[\prod_{(i,j)} R_{(i,j)}\Big]^r \to e^{-ist}e^{-itA_{\text{hit}}} = e^{+iA_1 t}$ as $r \to \infty$ ($t = \pi/s$)

package as built: $r = 1$; 4.7 prices $r = 4$ (error 1.48 at $r = 1$, 0.43 at $r = 4$, 7.13)

(b) One pair block c-$R_{(0,9)}$ written out ($n = 4$; $|0\rangle = |0000\rangle$, $|9\rangle = |1001\rangle$, bits $h_3 \dots h_0$)



1. Pivot h_0 (lowest differing bit); CNOT $h_0 \to h_3$: $|9\rangle \to |1\rangle = |0001\rangle$, $|0\rangle \to |0\rangle$.

2. X on h_1, h_2, h_3 (zero in $|1\rangle$), so the pair reads $|111\,x\rangle$.

3. $R_X(2t/r)$ on h_0, controlled by T, h_1, h_2, h_3: acts on $\text{span}\{|0\rangle, |9\rangle\}$ only.

4. Undo 2 and 1. The same block shape as the segment case (C2b); only the operator's edges differ. A clique has $m(m-1)/2$ edges, all overlapping, so the terms do not commute and $r > 1$ is needed.